

Summary

1. Relaxation of architecture, to bilinear forms
2. Effective loss of fixed embedding under perturbative NN interaction that breaks the perm. sym.
3. optimize the loss, find that the shift is weighted avg of other embeddings, weighted by bigram stats
4. open questions
 - (1) higher order expansion & non-perturbative causes of preferred n-grams?
 - (2) current progress as 'write down tautological interaction terms that depends on the Gaussian theory. then derive corrections to embedding
↳ this physical?
 - (3) How to get bigram stats from the Sequence joint parts?

$$\lambda \ll 1 \text{ if } G_{ij} \sim O(1)$$

$$p(t_1, \dots, t_L) = e^{-\left(\underbrace{\sum_{i,j} G_{ij} n_i n_j}_{\text{as if iid.}} + \lambda \sum_{i=1}^{L-1} e_i \cdot e_{i+1} \right)}$$

Q: an architecture that learns this, asym. ↓

old: set $e_i \cdot e_j = \Sigma_{ij}$,

what would embedding
change if we insist on
the architecture
remains unchanged.

affine layer $\Lambda_r^{-1}(e_{w_t} - \bar{e}) = \underline{z_t} \rightarrow$
↓ mean of emb.

sum & square layer.
(quadratic form)

$$W = \mathbb{1} + \lambda \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$$

input independent?



$$\underbrace{(z_1, z_2, \dots, z_L)^T \begin{pmatrix} W \end{pmatrix} \begin{pmatrix} z_1 \\ \vdots \\ z_L \end{pmatrix} + \begin{pmatrix} b \end{pmatrix} \begin{pmatrix} z_1 \\ \vdots \\ z_L \end{pmatrix}}_{U^T S U}$$

$$\underline{e_i} = U \sqrt{\Lambda}$$

$$\begin{aligned} \underline{z_i} \cdot \underline{z_i} &= \Lambda_r^{-1/2} (e_{w_i} - \bar{e}) \cdot (e_{w_i} - \bar{e}) \\ &= \Lambda_r^{-1/2} e_{w_i} \cdot e_{w_i} - e_{w_i} \cdot \bar{e} \end{aligned}$$

$$\underline{e_i} \cdot \underline{e_j} = \Sigma_{ij}$$

$$\underline{z_t} = \Lambda_r^{-1} (e_{w_t} - \bar{e})$$

↪ diagonal

$$\log p = c + \underline{z}^T W \underline{z}, \quad W = \mathbb{1}$$

$$= \underline{e_{w_1}} \cdot \sum p_i \underline{e_i} \quad \left[\underline{\sum p_i} \quad \underline{\sum w_i} \right]$$

for each token.

$$\downarrow \quad \underline{z} \mapsto \sqrt{S} U \underline{z}$$

$$\boxed{\sum_{i=1}^N n_i = 0} \rightarrow$$

make this exact!

$$\underline{z_i'}^T \underline{z_t'} = \underline{z}^T$$

linear transform of $\underline{z}$

linear mixing of $e_i \mapsto e_i'$

Details . $P(t_1, t_2, \dots, t_L) = e^{-G_{ij} n_i n_j + \epsilon_{NN}}$

$$\log P(t_1, \dots, t_L) \propto - \left(\sum_{i,j} e^{-G_{ij} n_i n_j} + \lambda \sum_{\mu=1}^L e_{t_\mu} \cdot e_{t_{\mu+1}} \right)$$

convert to summation over μ

$$n_i = \left(\sum_{\mu=1}^L \delta_{t_\mu i} - p_i \right)$$

$$\sum_{i,j} G_{ij} \left(\sum_{\mu=1}^L \delta_{t_\mu i} - p_i \right) \left(\sum_{\nu=1}^L \delta_{t_\nu j} - p_j \right)$$

(mean freq. deduction)

$$= \underbrace{\sum_{\mu, \nu} (G_{t_\mu t_\nu})}_{\text{Tr } G^2} - \underbrace{\sum_{i,j, \mu} G_{ij} \delta_{t_\mu i} p_j}_{\sum_{j, \mu} G_{t_\mu j} p_j} - \sum_{i,j, \nu} G_{ij} p_i + \underbrace{\sum_{i,j} G_{ij} p_i p_j}_{\text{Inverse of PMI?}}$$

$\text{Tr } G^2$

$$\sum_{j, \mu} G_{t_\mu j} p_j$$

$$\sum_{i, \mu} G_{i t_\mu} p_i$$

$$\log P(t_1, \dots, t_L) \propto - \text{Tr } G^2 + \lambda \text{Tr}(\underline{E^T \Lambda_{+1} E})$$

Generalized Architecture:

① embed. $t_\mu = i \mapsto v_i$

COM

② affine map $z_t = \Lambda_r^{-1} (e_{w_t} - \bar{e})$ mixing in ctx!

③ bilinear product.

$$\hat{Q} = z^T W z + \cancel{b}$$

ctx-by-ctx.

$$W = \mathbf{1} + \lambda \cdot \Lambda_{+1}$$

"shift-by-one" matrix.

$$z = (\text{ctx}, d_m)$$

$$= \begin{pmatrix} | & | & & | \\ z_1 & z_2 & \dots & z_L \\ | & | & & | \end{pmatrix}$$

$$\hat{Q} = \text{Tr}(z^T z) \xrightarrow{\text{reduce}} \text{Tr}(z^T W z)$$

Polar Decomposition of $W = \mathbf{1} + \lambda \Lambda_{+1}$ gives.

$$W = U_W^T \sqrt{S_W} \sqrt{S_W} U_W$$

Incorporate $z \mapsto (\sqrt{S_W}^{-1} U_W^T z)$
ctx-by-ctx.

want an embedding $\{e'_i\}$ such that

mix ctx. also.

$$z'_t = \Lambda_r^{-1} (e'_{w_t} - \bar{e}) \stackrel{!}{=} \sqrt{S_W}^{-1} (U_W)^T \Lambda_r^{-1} (e_{w_t} - \bar{e})$$

$$\sqrt{W} : (\text{ctx}, \text{ctx})$$

T.T

N vectors, one for each $w_t = 1, 2, \dots, N$

$$\Rightarrow \boxed{e'_{w_t} - \bar{e}' \stackrel{!}{=} S_w^{-1/2} U_w^T (e_{w_t} - \bar{e})}$$

(ctx, ctx)

$$\begin{cases} e_{w_t} - \bar{e} = e_{w_t} - \sum_{i=1}^N p_i e_i \\ e'_{w_t} - \bar{e}' = e'_{w_t} - \sum_{i=1}^N p_i e'_i \rightarrow \end{cases}$$

but we only have freedom for $e_{w_t} \mapsto e'_{w_t}$

group eqns for the same i
minimising the loss

$$\begin{aligned} \mathcal{L} &= \sum_{t=1}^L \| \tilde{e}'_{w_t} - S_w^{-1/2} U_w^T \tilde{e}_{w_t} \| \\ &= \sum_{i=1}^N (p_i + n_i) \| e'_i - S_w^{-1/2} U_w^T \end{aligned}$$

same avg of many.

possibly inconsistent, U_w
vectors.

want a set of e_i @
 $\mathcal{L}$ minimiser, written in terms of
the perturbations

$$\mathcal{F} = (G_{ij} n_i n_j) + \lambda \sum_{t=1}^L (e_{w_t} - \bar{e}) \cdot (e_{w_{t+1}} - \bar{e})$$

ideal: $w_t \rightarrow e_{w_t} \Rightarrow z_t = \Lambda^{-1} (e_{w_t} - \bar{e}) \Rightarrow \log P = \text{Tr}(\underline{z}^T \underline{W} \underline{z})$

$$\boxed{W = \mathbb{1} + \lambda T_{-1}} \quad \rightarrow \quad e'_{w_t} \Rightarrow z'_t = \Lambda^{-1} (e'_{w_t} - \bar{e}') \Rightarrow \log P = \text{Tr}(\underline{z}'^T \underline{z})$$

same map @ any pos.

$$(\mathbb{1} + \lambda T)^{1/2} = \mathbb{1} + \frac{1}{2} \lambda T$$

$$\boxed{z' = \sqrt{W} z} \quad \rightarrow \quad \text{mixing ctx positions}$$

$$e_{w_t} \rightarrow \sqrt{W} e_{w_t}$$

propose e'_{w_t} such that $e'_{w_t} = \underset{\{e_{w_t}\}}{\text{argmin}} \sum_{t=1}^L \| e'_{w_t} - \sqrt{W} e_{w_t} \|^2$

$$e'_{w_t} = \underset{\{e_{w_t}\}}{\text{argmin}} \sum_{i=1}^{d_v} (p_i + n_i) \| e'_{w_t} - \sqrt{W} e_{w_t} \|^2$$

desired embedding matrix = mixture of current embeddings.

$$\underline{e_{w_t}} \rightarrow \underline{\sqrt{W}} e_{w_t} \stackrel{Q}{=}$$

$$X: (\text{ctx}, \text{model}) \rightarrow (\text{ctx}, \text{ctx}) \cdot (\text{ctx}, \text{model})$$

$$e_{w_t} \rightarrow Q_{w_t, w_t'} e_{w_t'}$$

$\{e_i'\}$ new set of embeddings.

$$L = \sum_{t=1}^L \|e_{w_t}' - Q_{t,t'} e_{w_t}\|$$

group terms involving token i , or, mapping: $\gamma(i) = \{t \mid w_t = i\}$, $|\gamma(i)| = p_i$

$$L = \sum_{i=1}^d \sum_{t \in \gamma(i)} \|e_i' - Q_{t,t'} e_{w_t}\|$$

$$L(e_i') = \mathbb{E}_{\{w_1, w_2, \dots, w_L\}} \sum_{i=1}^d \sum_{t \in \gamma(i)} \|e_i' - Q_{t,t'} e_{w_t}\|$$

(fixed.)

$(e_{w_t} + e_{w_{t+1}}, \frac{\lambda}{2})$

bigrams.

$$L(e_i') = \sum_{\{w_1, \dots, w_L\}} e^{-G_{ij} n_i n_j + \sum e_{w_t} \cdot e_{w_{t+1}}} \sum_{i=1}^d \sum_{t \in \gamma(i)} \left\| \delta e_i - \frac{\lambda}{2} e_{w_{t+1}} \right\|^2$$

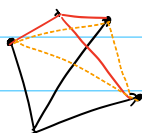
look at all appearances, & bigrams.

$$\text{then } \sum_{t \in \gamma(i)} \left\| \delta e_i - \frac{\lambda}{2} e_{w_{t+1}} \right\|^2$$

given bigram stats. $p(j|i)$. $p(w_{t+1} = j \mid w_t = i)$

$$[L(p_i + n_i)] \sum p(w_{t+1} = j \mid w_t = i) \left\| \delta e_i - \frac{\lambda}{2} e_j \right\|^2$$

$$\sum_{i=1}^d L(p_i + n_i) \sum p_{ji}^{(2)} \left\| \delta e_i - \hat{e}_j \right\|^2$$



$$L = \sum \lambda_i \|v - v_i\|^2$$

→ New embedding to 1st order in λ :

$$\frac{\partial L}{\partial v} = \sum \lambda_i (\|v\|^2 - 2v \cdot v_i) + \cancel{\|v_i\|^2}$$

$$= \sum \lambda_i (2v - 2v_i)$$

$$\rightarrow \sum \lambda_i (\vec{v} - \vec{v}_i) = 0$$

$$\sum_j P_{ij} (\vec{ae}_i - \vec{e}_j) = 0$$

$$\sum_j P_{ij} \vec{ae}_i - P_{ij} \vec{e}_j$$

$$\vec{ae}_i = \sum_j P_{ij} \vec{e}_j$$

How can P_{ij} be calculated from the Gaussian Theory.

(bigram stats)

$$P(w_1, w_2, \dots, w_L) = e^{-\left(G_{ij} n_{ij} + \lambda \sum_{t=1}^L (e_{w_{t+1}} \cdot e_{w_t})\right)}$$

$$P(w_{t+1} = j | w_t = i) = \frac{P(w_{t+1} = j, w_t = i)}{P(w_t = i)} \quad (P_i)$$

$$P(w_{t+1} = j | w_t = i) \approx e^{-\lambda e_i \cdot e_j}$$